KARTHIKEYAN P

AI · IoT & Embedded Systems Developer | B.Tech CSE, Chandigarh University | Patent Holder | Research Intern — IIT Guwahati & NCU &CGU Taiwan

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Technical Skills

Languages: Python, C++, JavaScript, Java, HTML/CSS
AI / ML: Scikit-Learn, TensorFlow, PyTorch, XGBoost, OpenCV, HuggingFace Transformers, TensorFlow Lite, NLP
IoT & Embedded: NodeMCU ESP32, Raspberry Pi, Arduino, ESP Modules, MEMS Sensors, PPG, RTOS, Bayesian Sensor Fusion
Web & Backend: React.js, Node.js, Flask, FastAPI, Express.js, MongoDB, MySQL, REST APIs, Blynk IoT
Tools: Git, GitHub, VS Code, Jupyter, Google Colab, Power BI, Linux, Vercel

Experience

Chang Gung University July 2026- Present
Researcher *Taoyuan, Taiwan*

- Joined as a Researcher at Chang Gung University, continuing applied AI, embedded systems, and human-robot interaction research. (Department of artificial intelligence)
- Contributing to the RobotNurseHelper initiative, developing a client-server based social-robot system (Kebbi Air-S / Zenbo Junior II) for patient education and nurse-support applications in Taiwanese hospitals.

Q2Q Solutions 2026 – Present
ERP Developer *India*

- Developing ERP modules to automate business workflows; collaborating with stakeholders to architect and deploy production-ready features.

National Central University — Center for General Education March – July 2026
Research Assistant *Taoyuan, Taiwan*

- Independently assembled an F450 quadcopter, performed flight-controller calibration, and built a Raspberry Pi–powered onboard image-analysis pipeline.
- Co-designing a cognitive science experiment on drone-operation skill acquisition using DJI drones and Garmin SDK toolchain.

Indian Institute of Technology, Guwahati July 2025
Summer Research Intern *Guwahati, India*

- Selected for a competitive summer internship at IIT Guwahati; worked on applied AI and engineering research projects.

Matt Engineering Solutions June – July 2025
Internship Trainee — IoT *India*

- Prototyped IoT-based solutions using Arduino, ESP modules, and sensor integration for embedded systems applications.

SMG Electric Scooters Ltd. Dec 2025 – Jan 2026
IT Intern *New Delhi, India*

- Supported IT operations in an EV manufacturing environment; delivered quality work consistently under deadlines.

EDIFICE Technologies 2025
Web Development Intern *India*

- Built and enhanced web application interfaces using HTML, CSS, and JavaScript within a full-stack development team.

Research & Publications

Patent: Multi-Sensor IoT Vehicle Safety System with Predictive Accident Prevention Filed Nov 2025
Indian Patents Act, 1970 — Filed via Chandigarh University *IoT / Automotive Safety*

- Designed a multi-sensor IoT safety platform (NodeMCU ESP32 + Raspberry Pi 4) integrating IR eye-tracking, alcohol, cardiac (PPG), MEMS collision, and flame sensors with a Bayesian fusion engine, achieving 95% detection accuracy and 60% fewer false positives than single-modal systems.
- Implemented a 5-stage automated intervention protocol with dual-channel (Wi-Fi + GSM) emergency coordination and V2I communication for smart city integration.

Research Paper — Intelligent Transportation Systems 2025
Published through Chandigarh University

- Authored a research paper presenting data-driven findings in IoT-based vehicle safety aligned with the patented system.

Projects

Drone Vision Research Platform | *Raspberry Pi, OpenCV, F450 Quadcopter, DJI SDK, Python* 2026

- Built and flew an F450 quadcopter from scratch; integrated Raspberry Pi for real-time onboard image analysis — combining embedded systems, CV, and drone hardware.
- Bridging hardware-software integration for a cognitive experiment studying drone-skill acquisition across working-memory and motor-reaction-time profiles.

RobotNurseHelper — Social Robot for Patient Education **2026**

Raspberry Pi, Android (Kebbi Air-S / Zenbo Junior II), OpenCV, NLP, Client-Server Architecture

- Collaborating on an interactive desktop robot system aimed at easing the shortage of nurse educators in Taiwanese hospitals, focused on delivering pre- and post-surgical health education to cataract patients.
- Working with compact social robots (Kebbi Air-S, Zenbo Junior II) in a client-server architecture that offloads computer-vision and NLP processing to a backend server, so the robot itself handles real-time speech and patient interaction.

PhishGuard AI | *Python, Scikit-Learn, XGBoost, Flask* 2025

- AI phishing URL detector using Random Forest + XGBoost ensemble with 30+ engineered features, achieving ~95% accuracy; served via Flask with real-time confidence scoring.

IoT Patent Prototype — Vehicle Safety System | *NodeMCU ESP32, Raspberry Pi 4, TensorFlow Lite, OpenCV, C++* 2025

- Hardware prototype of the patented system: multi-sensor fusion, 5-stage safety intervention, and live GPS + vitals transmission to Blynk IoT cloud dashboard via Wi-Fi/GSM.

ML & AI Projects Collection | *Python, TensorFlow, PyTorch, Scikit-Learn, HuggingFace* 2024 – 2025

- CNN (CIFAR-10), RNN/LSTM, Transformer experiments; NLP pipelines with BERT fine-tuning and spaCy for text classification and NER.

Education

Chandigarh University *(QS - #575)* 2024 – 2028

B.Tech, Computer Science & Engineering | ***CGPA: 8.15/10*** *Ludhiana, Punjab, India*

National Central University (Semester Exchange) *(QS - #587)* 2026

School of Engineering, Computer Science & Information Engineering | ***CGPA:4.22/4.3*** *Taiwan*

Leadership & Extracurricular

Campus Ambassador — Google Gemini July 2026 – Present

Google

- Serving as a Campus Ambassador for Google Gemini, promoting AI literacy and driving student engagement with Gemini tools on campus**.**

University-Level Hackathon Winner — AI & Web Development Track 2024 – 2025

Chandigarh University

- Won multiple university hackathons leading cross-functional teams through rapid prototyping and live demos before industry judges.

Class Representative, B.Tech CSE — Chandigarh University Aug 2024 – Jul 2025

- Liaison between student cohort and faculty; organised study initiatives and coordinated academic scheduling

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